

Class 8: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class. You'll extend what you've learned by combining PCA as a preprocessing step to clustering using data that consist of measurements of cell nuclei of human breast masses. This expands on our RNA-Seq analysis from last day.

The data itself comes from the Wisconsin Breast Cancer Diagnostic Data Set first reported by K. P. Benne and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets".

Values in this data set describe characteristics of the cell nuclei present in digitized images of a fine needle aspiration (FNA) of a breast mass.

Import

Data was downloaded from the class website as a CSV file.

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names=1)
head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
84348301	0.14250	0.28390	0.2414	0.10520	
84358402	0.10030	0.13280	0.1980	0.10430	
843786	0.12780	0.17000	0.1578	0.08089	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
84358402	0.1809	0.05883	0.7572	0.7813	5.438
843786	0.2087	0.07613	0.3345	0.8902	2.217
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
84358402	0.01756	0.005115	22.54	16.67	
843786	0.02165	0.005082	15.47	23.75	

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249

	concavity_worst	concave.points_worst	symmetry_worst
842302	0.7119	0.2654	0.4601
842517	0.2416	0.1860	0.2750
84300903	0.4504	0.2430	0.3613
84348301	0.6869	0.2575	0.6638
84358402	0.4000	0.1625	0.2364
843786	0.5355	0.1741	0.3985

	fractal_dimension_worst
842302	0.11890
842517	0.08902
84300903	0.08758
84348301	0.17300
84358402	0.07678
843786	0.12440

Data Exploration

The first column `diagnosis` is the expert opinion on the samep (i.e. patient FNA)

```
head(wisc.df$diagnosis)
```

```
[1] "M" "M" "M" "M" "M" "M"
```

Remove the diagnosis from data for subsequent analysis

```
wisc.data <- wisc.df[,-1]
dim(wisc.data)
```

```
[1] 569 30
```

Store the diagnosis as a vector for use later when we compare our results to those from experts in the field.

```
diagnosis <- factor(wisc.df$diagnosis)
```

Q1. How many observations are in this dataset?

There are 569 observations/patients in the dataset

Q2. How many of the observations have a malignant diagnosis?

```
diagnosis <- as.factor(wisc.df$diagnosis)
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#colnames(wisc.data)
length(grep("_mean", colnames(wisc.data)) )
```

```
[1] 10
```

Principal Component Analysis (PCA)

The `prompt()` function to do PCA has a `scale=FALSE` default. In general we nearly always want to set this to `TRUE` so our analysis is not dominated by columns/variables in our dataset that have high standard deviations and mean when compared to others just because the units of measurements are on different units/scales.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28

Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

```
pve <- (wisc.pr$sdev^2) / sum(wisc.pr$sdev^2)
pve[1] # variance explained by PC1
```

```
[1] 0.4427203
```

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

```
which(cumsum(pve) >= 0.70)[1]
```

```
[1] 3
```

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

```
which(cumsum(pve) >= 0.90)[1]
```

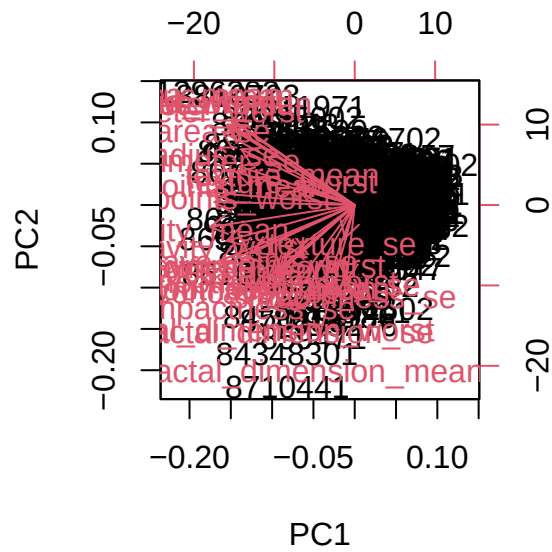
```
[1] 7
```

Interpreting PCA results

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The biplot is cluttered and difficult to interpret visually because too many things overlap.

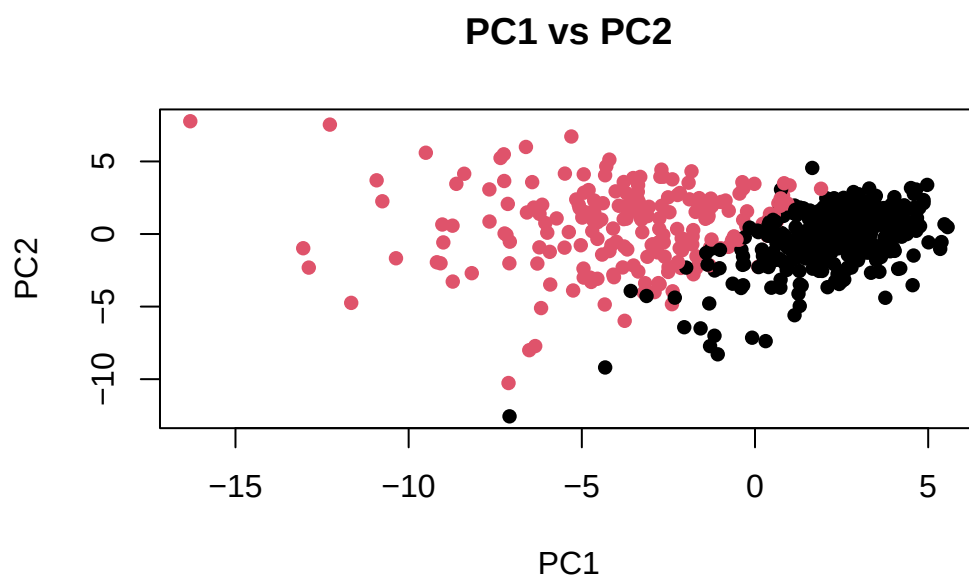
```
biplot(wisc.pr)
```



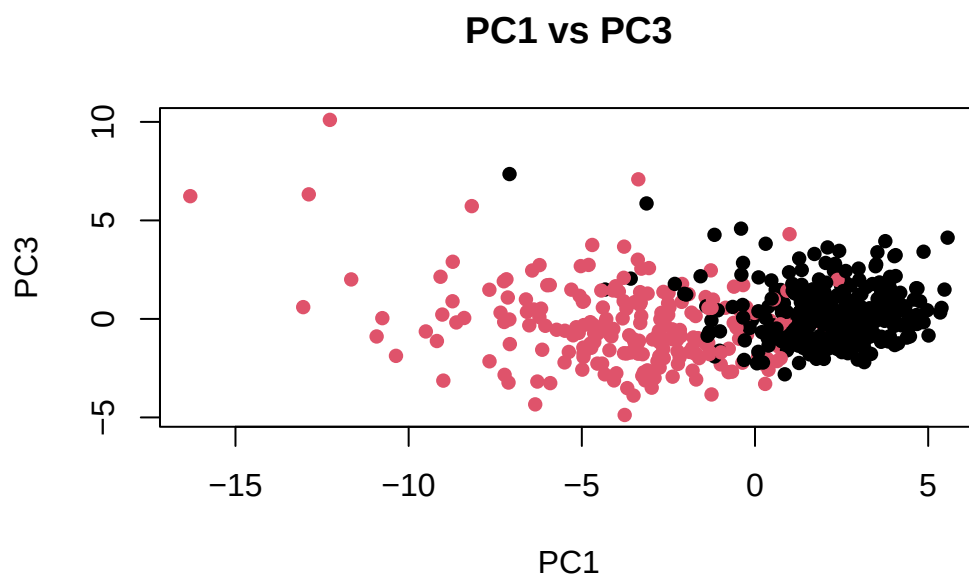
Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

I notice that PC1 clearly separates malignant(M) from benign(B). Additionally PC2 and PC3 contribute less to the separation.

```
plot(wisc.pr$x[,1], wisc.pr$x[,2], col = diagnosis,
     xlab = "PC1", ylab = "PC2", pch = 16, main = "PC1 vs PC2")
```



```
plot(wisc.pr$x[,1], wisc.pr$x[,3], col = diagnosis,  
     xlab = "PC1", ylab = "PC3", pch = 16, main = "PC1 vs PC3")
```

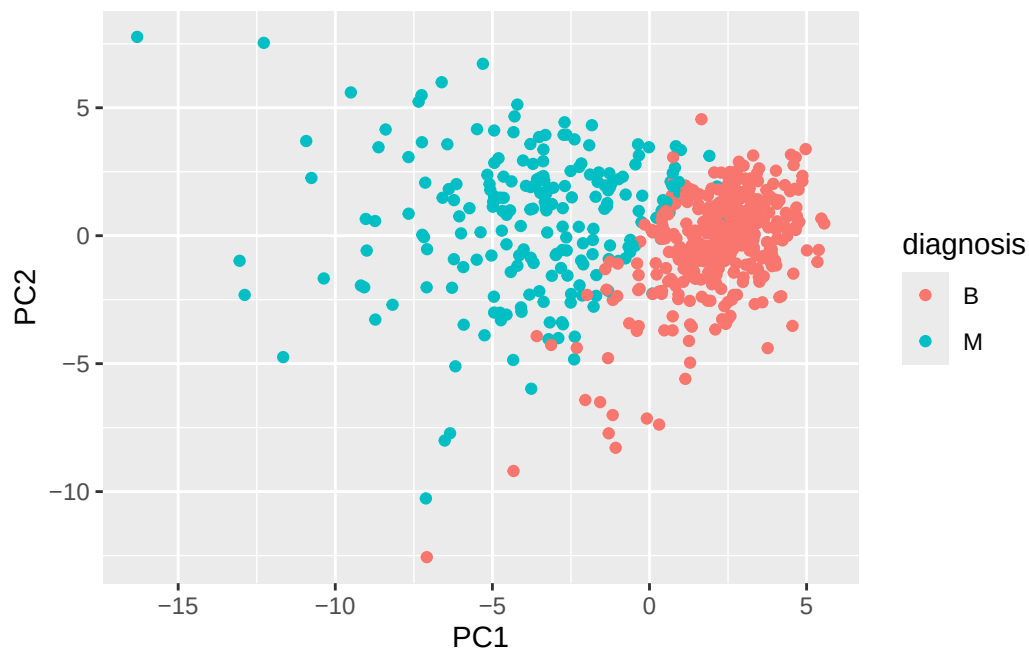


PCA Score Plot

The main PC result figure is called a “score plot” or “PC plot” or “ordination plot”...

```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



PCA Screen-Plot

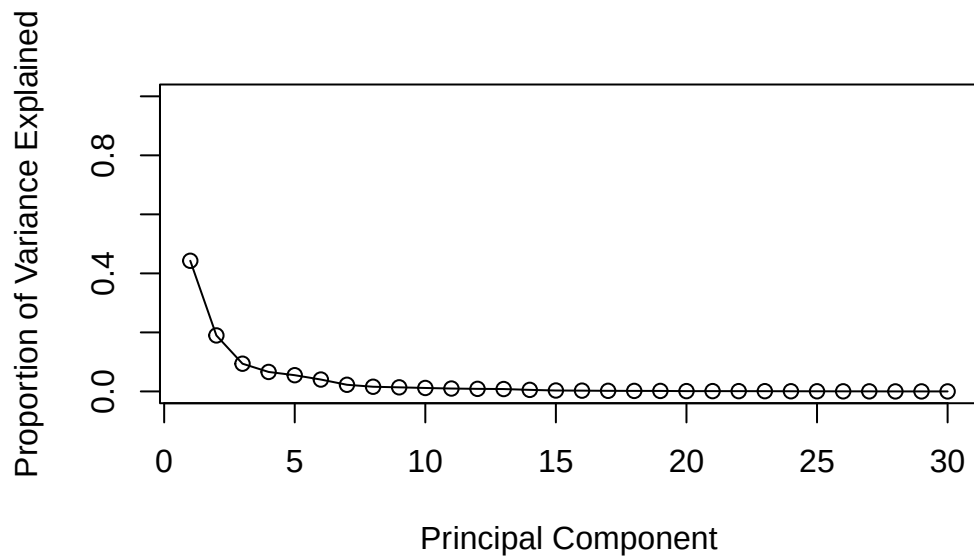
A plot of how much variance each PC captures.

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

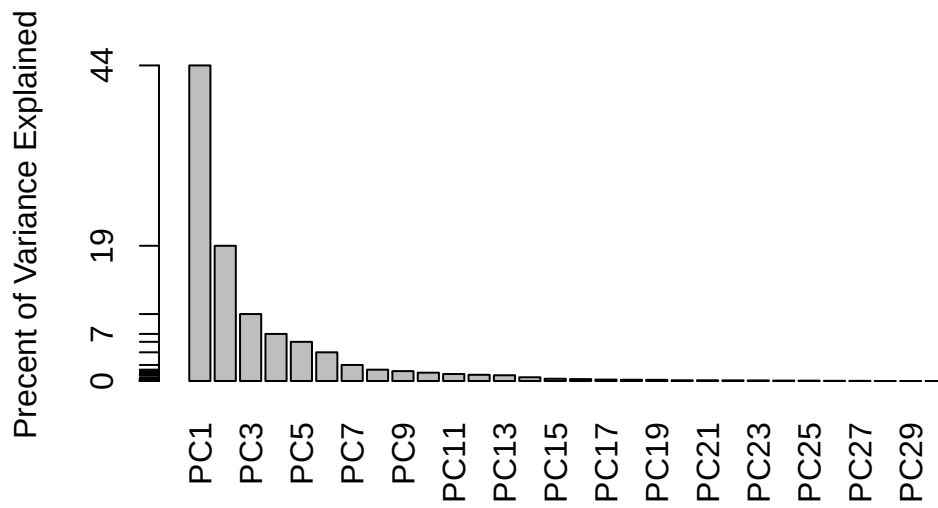
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```



```
pve <- pr.var/sum(pr.var)
plot(pve, xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
barplot(pve, ylab = "Precent of Variance Explained",
       names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`?

```
wisc.pr$rotation["concave.points_mean", "PC1"]
```

```
[1] -0.2608538
```

Q10. What is the minimum number of principal components required to explain 80% of the variance of the data?

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624

Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Hierachical clustering

Just clustering the original data is not very informative or helpful

```
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist)
```

View the clustering dendrogram result

```
plot(wisc.hclust)
```

Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

Q11. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters )
```

```
wisc.hclust.clusters
  1  2  3  4
177  7 383  2
```

```
table(wisc.hclust.clusters, diagnosis)
```

```

          diagnosis
wisc.hclust.clusters  B  M
1      12 165
2       2   5
3     343  40
4       0   2
```

Q12. Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 10?

```

for (k in 2:10) {
  cat("\nClusters:", k, "\n")
  print(table(cutree(wisc.hclust, k), diagnosis))
}

```

```

Clusters: 2
  diagnosis
    B    M
1 357 210
2   0   2

```

```

Clusters: 3
  diagnosis
    B    M
1 355 205
2   2   5
3   0   2

```

```

Clusters: 4
  diagnosis
    B    M
1  12 165
2   2   5
3 343  40
4   0   2

```

```

Clusters: 5
  diagnosis
    B    M
1  12 165
2   0   5
3 343  40
4   2   0
5   0   2

```

```

Clusters: 6
  diagnosis
    B    M
1  12 165
2   0   5
3 331  39

```

4	2	0
5	12	1
6	0	2

Clusters: 7

diagnosis		
	B	M
1	12	165
2	0	3
3	331	39
4	2	0
5	12	1
6	0	2
7	0	2

Clusters: 8

diagnosis		
	B	M
1	12	86
2	0	79
3	0	3
4	331	39
5	2	0
6	12	1
7	0	2
8	0	2

Clusters: 9

diagnosis		
	B	M
1	12	86
2	0	79
3	0	3
4	331	39
5	2	0
6	12	0
7	0	2
8	0	2
9	0	1

Clusters: 10

diagnosis		
	B	M

1	12	86
2	0	59
3	0	3
4	331	39
5	0	20
6	2	0
7	12	0
8	0	2
9	0	2
10	0	1

Q13. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

ward.D2 usually gives the clearest separation.

```
methods <- c("single", "complete", "average", "ward.D2")
for (m in methods) {
  cat("\nMethod:", m, "\n")
  hc <- hclust(data.dist, method = m)
  print(table(cutree(hc, 4), diagnosis))
}
```

Method: single

	diagnosis	
	B	M
1	356	209
2	1	0
3	0	2
4	0	1

Method: complete

	diagnosis	
	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Method: average

	diagnosis	
	B	M

1	355	209
2	2	0
3	0	1
4	0	2

Method: ward.D2

	diagnosis	
	B	M
1	0	115
2	6	48
3	337	48
4	14	1

Combining methods (PCA and Clustering)

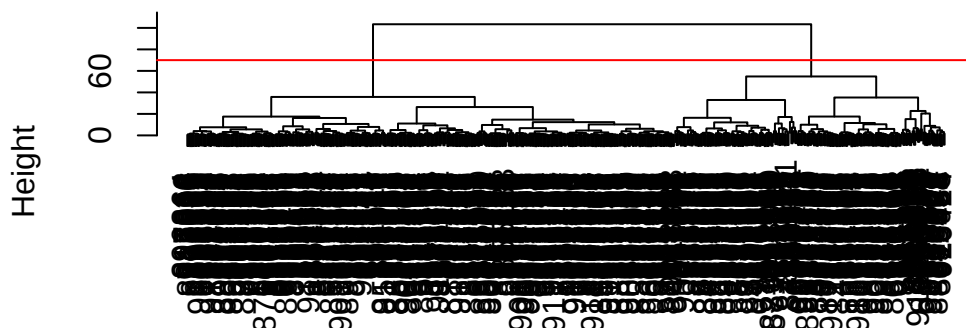
Clustering the original data was not very productive. The PCA results looked promising. Here we combine these methods by clustering from our PCA results. In other words “clustering in PC space”...

```
## Take the first 3 PCs
dist.pc <- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(dist.pc, method = "ward.D2")
```

view the tree...

```
plot(wisc.pr.hclust)
abline(h=70, col="red")
```


Cluster Dendrogram



```
dist.pc
hclust (*, "ward.D2")
```

To get our clustering membership vector (i.e. our main clustering result) we “cut” the tree at a desired height or to yield a desired number of “k” groups.

```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1  2
203 366
```

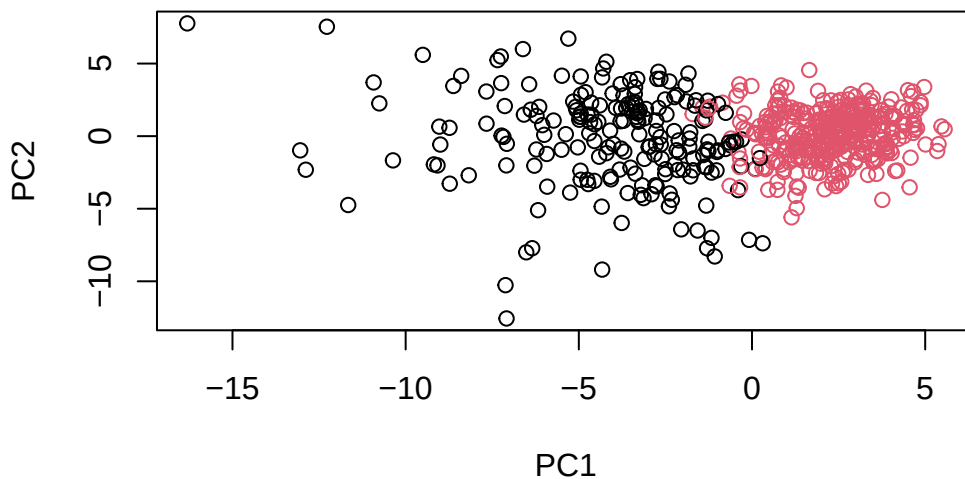
How does this clustering grps compare to the expert diagnosis

Q15. How well does the newly created model with four clusters separate out the two diagnoses?

```
table(grps, diagnosis)
```

```
diagnosis
grps    B    M
  1   24 179
  2  333  33
```

```
plot(wisc.pr$x[,1:2], col=grps)
```



Q16. How well do the k-means and hierarchical clustering models you created in previous sections (i.e. before PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.km$cluster` and `wisc.hclust.clusters`) with the vector containing the actual diagnoses.

```
wisc.km <- kmeans(data.scaled, centers = 2, nstart = 20)
table(wisc.km$cluster, diagnosis)
```

```
diagnosis
  B   M
1 14 175
2 343  37
```

```
table(wisc.km$cluster, diagnosis)
```

```
diagnosis
  B   M
1 14 175
2 343  37
```

```
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B  M
1      12 165
2       2   5
3     343  40
4       0   2

```

Q17. Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

Sensitivity: $TP/(TP+FN)$ Specificity: $TN/(TN+FN)$

7. Prediction

We can use our PCA model for prediction with new input patient samples.

```

new <- read.csv("https://tinyurl.com/new-samples-CSV")
npc <- predict(wisc.pr, newdata = new)
npc[, 1:5] # show first few PC values

```

```

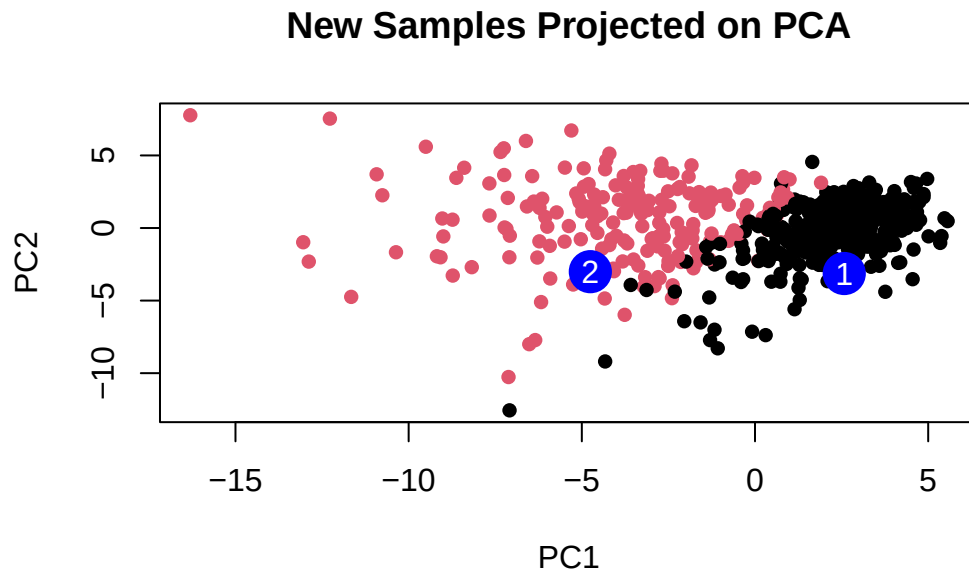
      PC1      PC2      PC3      PC4      PC5
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698

```

```

plot(wisc.pr$x[,1:2], col = diagnosis, pch = 16,
main = "New Samples Projected on PCA", xlab = "PC1", ylab = "PC2")
points(npc[,1], npc[,2], col = "blue", pch = 16, cex = 3)
text(npc[,1], npc[,2], labels = c("1", "2"), col = "white")

```



Q18. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize the new patient that is closer to the malignant cluster in PC space for follow up.